

Attempt 1



In Progress

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**Unlimited Attempts Allowed**

1/13/2025 to 4/25/2025

## ▼ Details

**P4: Two-Body Problem**

Consider two masses,  $m_1$  and  $m_2$ , under mutual gravitational influence. The position vectors for the two masses are  $\mathbf{r}_1$  and  $\mathbf{r}_2$ . From Newton's second law of gravitation, we get:

$$m_1 \cdot \mathbf{r}_1'' = \mathbf{F}, \quad m_2 \cdot \mathbf{r}_2'' = -\mathbf{F}, \quad \text{where } \mathbf{F} = -G \cdot m_1 \cdot m_2 \cdot (\mathbf{r}_1 - \mathbf{r}_2) / |\mathbf{r}_1 - \mathbf{r}_2|^3$$

where  $G$  is the gravitational constant.  $\mathbf{r}_1''$  and  $\mathbf{r}_2''$  are second derivatives of  $\mathbf{r}_1$  and  $\mathbf{r}_2$  with respect to time.

The two-body problem can be reduced to a one-body problem by a change of coordinates. Let  $\mathbf{r}$  be the relative position vector, and  $\mathbf{R}$  be the coordinate of the system; that is

$$\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2, \text{ and } \mathbf{R} = (m_1 \cdot \mathbf{r}_1 + m_2 \cdot \mathbf{r}_2) / (m_1 + m_2)$$

Now, changing coordinates in Newton's equations from  $\mathbf{r}_1$  and  $\mathbf{r}_2$  to  $\mathbf{r}$  and  $\mathbf{R}$  results in

$$m_1 \cdot \mathbf{R}'' + (m_1 \cdot m_2 \cdot \mathbf{r}) / (m_1 + m_2) = \mathbf{F}, \quad m_2 \cdot \mathbf{R}'' - (m_1 \cdot m_2 \cdot \mathbf{r}) / (m_1 + m_2) = -\mathbf{F}$$

and adding and subtracting these two equations yields the result

$$\mathbf{R}'' = 0 \text{ and } \mu \cdot \mathbf{r}'' = \mathbf{F}$$

where  $\mu$  is called the reduced mass and is given by  $\mu = (m_1 \cdot m_2) / (m_1 + m_2)$ .

The center-of-mass coordinate system is an inertial coordinate system (one moving at constant velocity with the center-of-mass fixed at the origin ( $\mathbf{R} = 0$ )). equation for the relative coordinate has the form of a one-body problem and can be written as

$$\mathbf{r}'' = -k \cdot \mathbf{r} / r^3, \text{ where } k = G (m_1 + m_2)$$

Once  $\mathbf{r}$  is determined, the coordinates of the two masses in the center-of-mass coordinate system can be found from

$$\mathbf{r}_1 = (m_2 \cdot \mathbf{r}) / (m_1 + m_2) \text{ and } \mathbf{r}_2 = - (m_1 \cdot \mathbf{r}) / (m_1 + m_2)$$

The equivalent one-body problem can be simplified further. The position vector  $\mathbf{r}$  and the velocity vector  $\mathbf{r}'$  form a plane. Since the acceleration vector points towards the position vector, the one-body motion will be restricted to this plane. There is no force component trying to move the masses off the plane. We can place a coordinate system so that  $\mathbf{r}$  lies in the x-y plane with  $z = 0$  and write  $\mathbf{r} = x \cdot \mathbf{i} + y \cdot \mathbf{j}$ .

The relevant units are length and time, and they may be nondimensionalized using the constant parameter  $k$  (which has units  $l^3 / t^2$ ) and the distance of the two masses (the minimum value of  $r$ ). The dimensionless governing equations, in component form, are then given by

$$x'' = (-x) / (x^2 + y^2)^{3/2} \text{ and } y'' = (-y) / (x^2 + y^2)^{3/2}$$

We can further orient the x-y axes so that the solution  $\mathbf{r} = \mathbf{r}(t)$  is symmetric about the x-axis, and the minimum value of  $r$  occurs at  $x = -1$  and  $y = 0$ , where requires  $x' = 0$ . The quantities  $x'$  and  $y'$  are the first derivatives with respect to time. Setting our initial conditions at the point of minimum  $r$ , we can write

$$x(0) = -1, y(0) = 0, x'(0) = 0, y'(0) = (1 + e)^{1/2}$$

where we have parameterized  $y'(0)$  using  $e$ . The parameter  $e$  is called the eccentricity of the orbit. Circular orbits have  $e = 0$ , and elliptical orbits have  $0 < e < 1$ ; for  $e = 1$ , the orbit is parabolic; for  $e > 1$ , we get a hyperbolic orbit. The period of a closed orbit is given by

$$T = 2\pi / (1 - e)^{3/2}$$

For this project, you will write code that solves the two-body problem. You will investigate the orbit for four mass ratios ( $m_1:m_2$ ) – 1:1, 1:2, 1:4, and 1:16. For each mass ratio, you will determine the orbit for the four eccentricity values – 0, 0.25, 0.50, and 0.75. Altogether, you will have 16 configurations.

You can write your own Runge-Kutta fourth-order solver or use the ODE solver in SciPy. Given the initial conditions, specified value of  $e$ , and mass ratio, solve the first-order differential equations for the sixteen configurations.

Browse papers in the Astrophysical Journal or Physical Review to get a feel of the writing style that we are expecting. You will be submitting a single paper. We have enclosed the LaTeX template that you have to use. Here is the outline of your paper:

- Title

- Authors' Names
- Affiliation Name (The University of Texas at Austin)
- Abstract
- Keywords
- In your introduction, state the physical situations in which the two-body problem arises and its scientific importance.
- Describe how you reduced the two-body problem to a one-body problem.
- State the initial conditions you chose and how you solved the equations.
- Display your solutions showing the orbits of the two bodies under the sixteen configurations.
- Summarize your findings.
- Discuss any future work you could have done to extend your work and contribute to our understanding of the two-body problem.
- Acknowledgment - Acknowledge any help that you got from your classmates or the TAs.
- References

Use the templates APJ\_Template.pdf and apj\_template.tex in Files-> Projects.

✓ View Rubric

| P4_Spring_2025                                                                               |                                                                                                                                                        |                 |
|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Criteria                                                                                     | Ratings                                                                                                                                                |                 |
| <b>Title</b><br><a href="#">view longer description</a>                                      | <b>Full Marks</b><br>Paper titled correctly<br>5 pts                                                                                                   | <b>No Marks</b> |
| <b>Author</b><br><a href="#">view longer description</a>                                     | <b>Full Marks</b><br>Authors mentioned clearly and correctly<br>2 pts                                                                                  | <b>No Marks</b> |
| <b>Affiliation</b><br><a href="#">view longer description</a>                                | <b>Full Marks</b><br>Affiliation mentioned correctly<br>5 pts                                                                                          | <b>No Marks</b> |
| <b>Abstract</b>                                                                              | <b>Full Marks</b><br>12 pts                                                                                                                            | <b>No Marks</b> |
| <b>Key</b>                                                                                   | <b>Full Marks</b><br>3 pts                                                                                                                             | <b>No Marks</b> |
| <b>Introduction</b><br><a href="#">view longer description</a>                               | <b>Full Marks</b><br>In your introduction, state the physical situations in which the two-body problem arises and its scientific importance.<br>10 pts | <b>No Marks</b> |
| <b>One body problem</b><br><a href="#">view longer description</a>                           | <b>Full Marks</b><br>Describe the reduction of the two-body problem to the one-body problem.<br>10 pts                                                 | <b>No Marks</b> |
| <b>Initial Conditions + Solving the equations</b><br><a href="#">view longer description</a> | <b>Full marks</b><br>State the initial conditions you chose and how you solved the equations.<br>10 pts                                                | <b>No marks</b> |

| Criteria                                                                        | Ratings                                                                                                                                                            |                                                             |                                                 |                                                                   |                                                    |                                                          |             |
|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------------|-------------|
| <b>Solution plots</b><br><a href="#">view longer description</a>                | <b>Fully correct</b><br><br>Display your solutions showing the orbits of the two bodies under the sixteen configurations.<br><br>20 pts                            | <b>Very close to the solution</b><br><br><br><br><br>18 pts | <b>Mostly correct</b><br><br><br><br><br>16 pts | <b>Some plots correct, some are off</b><br><br><br><br><br>12 pts | <b>Significant errors</b><br><br><br><br><br>8 pts | <b>Plots are not compatible</b><br><br><br><br><br>5 pts | <b>Zero</b> |
| <b>Summarization</b><br><a href="#">view longer description</a>                 | <b>Full Marks</b><br><br>Summarize your findings.<br><br>10 pts                                                                                                    |                                                             |                                                 |                                                                   | <b>No Marks</b>                                    |                                                          |             |
| <b>Conclusion</b><br><a href="#">view longer description</a>                    | <b>Full Marks</b><br><br>Discuss any future work you could have done to extend your work and contribute to our understanding of the two-body problem.<br><br>8 pts |                                                             | <b>Partial Marks</b><br><br><br><br>5 pts       |                                                                   |                                                    | <b>No Marks</b>                                          |             |
| <b>Acknowledgment and References</b><br><a href="#">view longer description</a> | <b>Full Marks</b><br><br>Acknowledge any help that you got from your classmates or the TAs.<br><br>5 pts                                                           |                                                             |                                                 |                                                                   | <b>No Marks</b>                                    |                                                          |             |



Availability Dates

1/13/2025 to 4/25/2025